National University of Computer and Emerging Sciences
Lahore Campus

Computer Organization & Assembly Language (EE2003)

Sessional-II Exam

Total Time (Hrs):	**1**
Total Marks:	**40**
Total Questions:	**3**

Date: November 4th 2025

Course Instructor(s)

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Roll No   Section   Student Signature

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Instruction/Notes: This is an open book exam. Only textbook by "Belal Hashmi" is allowed. Sharing book or calculators is NOT ALLOWED. PAPERS/NOTES ATTACHED WITH BOOK ARE NOT ALLOWED. Rough sheets can be used but will not be collected and checked. In case of any ambiguity, make reasonable assumptions. Questions during exams are not allowed.

CLO # 2: Describe the working of important x86 assembly primitives, including arithmetic, branching, bit manipulation, addressing modes and interrupt handling.

Q1: Short Questions [3+2+7+8 marks]

(a) Given that initial value of SP is 0x0000, what will be the value of SP after execution of the following instructions? Show working to get full credit.

push ax ret 3	

SP = 0x0003

3 Marks

(b) What will be the value of IP after execution of line 5 of the following code?

Line	offset	Code
1	0100	Call F1
2	0103	ret
3	0104	F1: mov ax, 0301
4	0107	Push ax
5	0108	Ret 2

IP = 0301 , 0x012D

2 Marks

0103

1

(c) The following subroutine tries to print '$' sign (black on white) at 4 different corners of the display screen. Identify and correct the errors in the code. You do not need to write the subroutine again. Just mention the line number and the code to be replaced.

drawChars:	Write your answer here.
1. mov ax, 0xb800	
2. mov cs, ax	mov es, ax
3. mov ah, '$'	mov ah, 0x70
4. mov al, 0x07	mov al, '$'
5. mov di, 0	
6. mov [es:di], ax	
7. mov di, 79	158
8. mov [es:di], ax	
9. mov di, 3838	3840
10. mov [es:di], ax	
11. mov di, 4000	3998
12. mov [es:di], ax	
13. ret	6x1 marks, ½ for correction ½ for marking

(d) Following code is trying to set bx=0 if data in the buffer and display memory is the same otherwise bx = 1. Your task is to complete the missing code (only).

```
[org 0x0100]
jmp start
buffer: times 4000 db 0
start: mov ax, 0xb800
mov ds, ax
push cs   pop es ; set ES here
mov di, buffer
mov si, 0
mov cx, 4000
repe cmpsb
cmp cx,0
je SameData
mov bx, 1
jmp exit
SameData: mov bx, 0
exit: mov ax, 0x4c00; terminate program
int 0x21
```

2 marks each

0453

CLO # 3: Apply the knowledge of Intel x86 architecture to develop moderately complex and well-modularized assembly programs.

Q2: [20 marks] Write a subroutine RemoveAndShift that takes a character and a row number as a parameter. The function should remove all the occurrences of that character from that row, and shifts the remaining characters on the row left (leaving spaces at the end). Credit will be given for using String Instructions where required. You can assume that the attribute byte is identical throughout the video memory.

E.g., The subroutine is passed the character 'a' and the row number 1. The initial screen is shown on the left. The subroutine removes all occurrences of character 'a' from 2nd row of the display (row number starts from 0) and shift the remaining characters to the left leaving spaces at the end as shown in the right screen. If the character is at the end of the row, you just need to replace it with the space.

Note that your code should be generic for any number of character occurrences in a row. The caller of the subroutine is already provided.

Sample Run: (This sample is for a display memory of 5x6 grid. Your code should be for 25x80 grid screen as done in class.)

a	b	c	d	e	f
b	r	a	t	a	s
c	v	b	r	u	g
q	e	w	d	e	w
w	e	r	q	a	r

a	b	c	d	e	f
b	r	t	s		
c	v	b	r	u	g
q	e	w	d	e	w
w	e	r	q	a	r

```
[org 0x100]
mov ah, 1; row number
mov al, 'a'; character
push ax
call RemoveAndShift
mov ax, 0x4C00
int 0x21

RemoveAndShift:
        push bp
        mov bp, sp
        pusha

        mov bx, [bp+4]
        mov bl, bh
        mov bh, 0            ; bx now contains row number
```

04

```
mov ax, 0xb800
mov es, ax
mov ax, 160
mul bx            ; calculating the value of di

mov di, ax
mov dx, ax
add dx, 158       ; dx pointing to the last cell of that row
mov ax, [bp+4]
mov si, 0
push si           ; si will count the loop iterations
mov ah, 0x07
push es
pop ds
mov cx, 80        ; no. of cells in that row

l1:
cld
repne scasw       ; finding for the character
mov si, di
mov bx, di
sub di, 2
push cx
rep movsw         ; moving the character
pop cx
add cx, 1
pop si
inc si
push si
mov di, bx
cmp di, dx
jnae l1
pop si
mov cx, si
mov di, dx
mov al, 0x20
std
rep stosw         ; printing the space at the end
popa
pop bp
ret 2
```